CHEMISTRY ADAPTLY

CHEMICAL CHANGES – OXIDATION & REDUCTION

KEY VOCABULARY

Oxidation: loss of electrons (or gain of oxygen).
Reduction: gain of electrons (or loss of oxygen).
Redox: a reaction where oxidation and reduction both happen.
OIL RIG: Oxidation Is Loss, Reduction Is Gain (of electrons).

COMMON MISCONCEPTIONS

~~Oxidation always involves oxygen.~~
✓ Oxidation is loss of electrons — oxygen need not be present.
~~Reduction means getting smaller.~~
✓ Reduction is gain of electrons / loss of oxygen.
~~Only one species changes in redox.~~
✓ One is oxidised AND another is reduced.

REDOX EXAMPLE

$Mg + Cl_2 \rightarrow MgCl_2$
$Mg \rightarrow Mg^{2+} + 2e^-$ (oxidation — loses e^-)
$Cl_2 + 2e^- \rightarrow 2Cl^-$ (reduction — gains e^-)
Electrons lost = electrons gained.
Use OIL RIG to decide which is which.

1 DEFINE (ELECTRONS)

Define **oxidation** and **reduction** in terms of electrons. **[2 marks]**

2 IDENTIFY

In $Zn \rightarrow Zn^{2+} + 2e^-$, state whether zinc is oxidised or reduced. **[1 mark]**

3 HALF EQUATION

Write the **half equation** for chlorine gaining electrons. **[2 marks]**

4 OXYGEN VIEW

Copper oxide reacts with hydrogen. **Explain** which substance is reduced. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

Oxidation is the _____ of electrons; _____ is the gain of electrons.

loss | gain | oxidation | reduction

6 REDOX OR NOT

Explain why a displacement reaction is a redox reaction. **[2 marks]**

TIP: *OIL RIG — Oxidation Is Loss, Reduction Is Gain (of electrons).*

✓ **Think: Why must reduction always happen at the same time as oxidation?**